

Verification & Validation Protocol

Document ID	VVP-MEC-001
Device	Six-Sense Diagnostic Microneedle Patch
Testing Phase	Design Verification (Mechanical Integrity)
Regulatory Alignment	21 CFR Part 820 (QMSR), ISO 13485:2016 Clause 7.3

1. Objective

To verify that the Six-Sense microneedle array withstands standard application forces without fracturing or undergoing critical deformation, satisfying design verification requirements under 21 CFR Part 820 (QMSR) and ISO 13485:2016 Clause 7.3.

2. Scope

This protocol applies to the simulated compressive load testing of 50 randomly sampled microneedle arrays across two manufacturing runs (Lot A and Lot B). This test evaluates the physical yield limit and structural integrity of the individual microneedles under direct axial load. Because the test is destructive, the evaluation is sample-based.

3. Equipment & Methodology

No consensus standard currently exists for compressive fracture testing of polymer microneedle arrays, so the following parameters are specified directly and held constant across all samples:

- **Equipment:** Simulated Instron universal testing machine with axial compression load cell.
- **Methodology:** A continuous downward compressive force is applied to the central axis of the patch at a constant displacement rate until physical failure.
- **Controlled Variables:** Number of needles loaded simultaneously, crosshead displacement rate, and compression angle are fixed and recorded, as published

microneedle compression data shows buckling load varies substantially with all three.

- **Data Capture:** Peak force (N) and displacement (mm) at the moment of structural fracture are recorded and exported to a CSV matrix.

4. Acceptance Criteria

Minimum Yield Stress Threshold: 120.0 MPa
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- To pass verification, the calculated engineering stress (force divided by cross-sectional area) at the moment of fracture must be ≥ 120.0 MPa.
- Units fracturing below this threshold are recorded as nonconforming. If a specific manufacturing lot demonstrates a high failure rate, it is flagged for mechanical root-cause analysis and corrective action per ISO 13485:2016 Clause 8.5.2.

5. Automated Analysis & Data Integrity

Consistent with ALCOA+ data integrity principles:

- Raw load-cell data is processed through an automated Python pipeline (Pandas / NumPy) that calculates engineering stress and strain, removing manual transcription steps between the source file and the reported values.
- The script tags units failing the acceptance criterion and generates a stress-strain scatter plot for visual review of lot-to-lot consistency.
- Because the analysis is scripted rather than hand-calculated, the same input file reproduces the same output, making the transition from raw data to verification decision traceable.